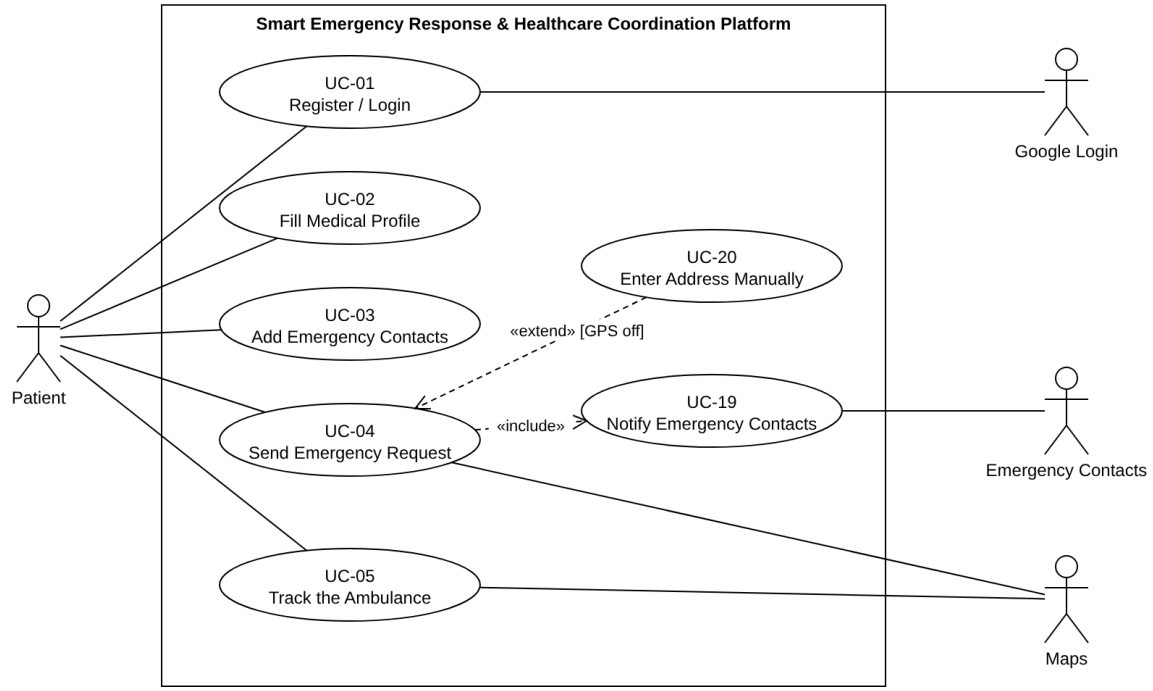
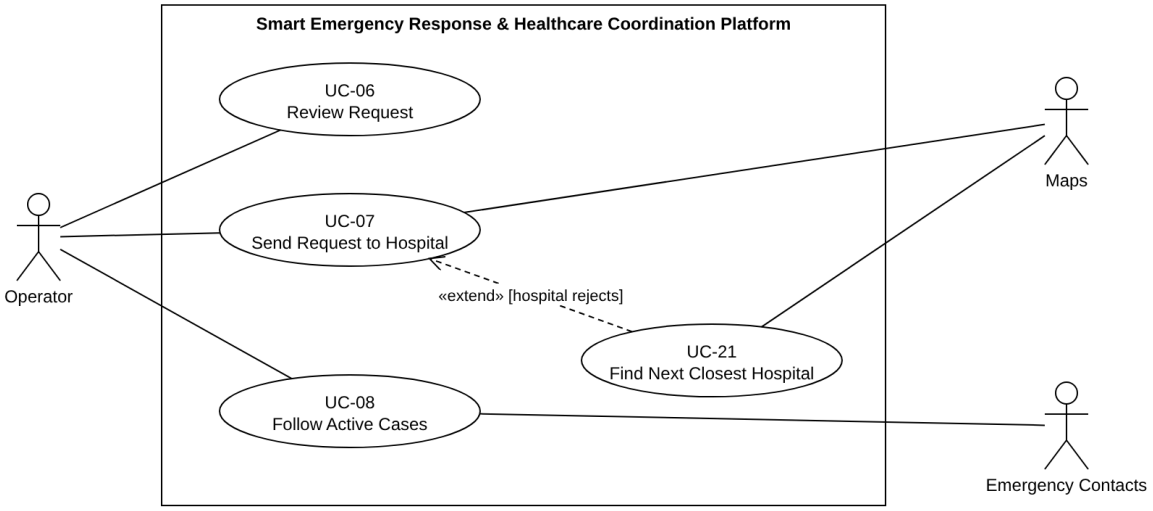
Use Case Descriptions

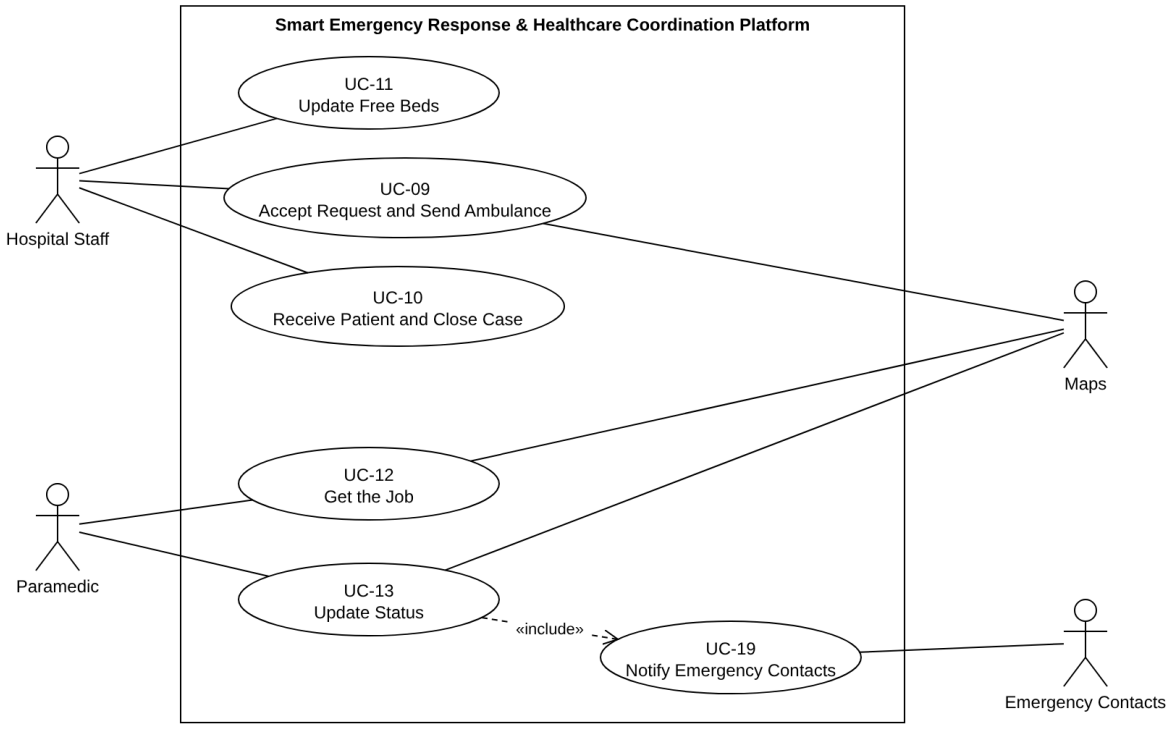
Smart Emergency Response & Healthcare Coordination Platform



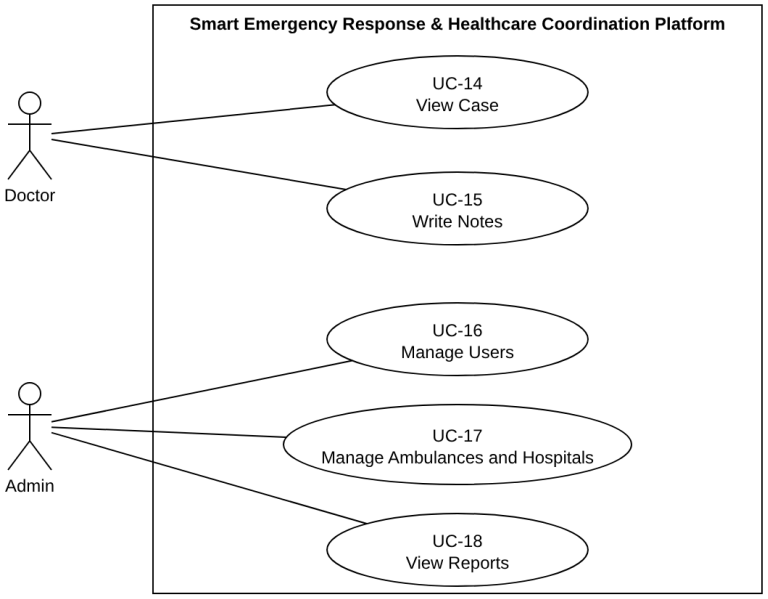
Use Case Diagram 1 - Patient



Use Case Diagram 2 - Operator



Use Case Diagram 3 - Hospital Staff and Paramedic



Use Case Diagram 4 - Doctor and Admin

Actors

Actor	What they do
Patient	sends the emergency request and follows it
Operator	checks the request, sends it to a hospital, and calls the patient or family if needed
Hospital Staff	accepts or rejects requests, sends the ambulance, receives the patient and closes the case
Paramedic	works with the hospital, gets the job, sees the patient's medical info and updates the status
Doctor	sees the case and writes notes
Admin	manages the system and sees reports

Other actors: they don't log in, but the system works with them.

Other actor	What it does
Google Login	lets the patient sign in with his Google account
Maps	gives the location, the map and the distance to ambulances and hospitals
Emergency Contacts	the patient's family, they get a message when he sends a request and when the case changes

Use case list

ID	Use case	Actor
UC-01	Register / Login	Patient
UC-02	Fill Medical Profile	Patient
UC-03	Add Emergency Contacts	Patient
UC-04	Send Emergency Request	Patient
UC-05	Track the Ambulance	Patient
UC-06	Review Request	Operator
UC-07	Send Request to Hospital	Operator
UC-08	Follow Active Cases	Operator
UC-09	Accept Request and Send Ambulance	Hospital Staff
UC-10	Receive Patient and Close Case	Hospital Staff
UC-11	Update Free Beds	Hospital Staff
UC-12	Get the Job	Paramedic
UC-13	Update Status	Paramedic
UC-14	View Case	Doctor
UC-15	Write Notes	Doctor
UC-16	Manage Users	Admin
UC-17	Manage Ambulances and Hospitals	Admin
UC-18	View Reports	Admin

UC-19	Notify Emergency Contacts	System (Emergency Contacts receive)
UC-20	Enter Address Manually	Patient
UC-21	Find Next Closest Hospital	Operator

Relationships

Relationship	From	To	Why
include	UC-04 Send Emergency Request	UC-19 Notify Emergency Contacts	The system always sends a message to the emergency contacts when the patient sends a request (FR-13).
include	UC-13 Update Status	UC-19 Notify Emergency Contacts	The contacts always get a short update when the status changes (FR-35).
extend	UC-20 Enter Address Manually	UC-04 Send Emergency Request	Only happens when GPS is off (FR-12).
extend	UC-21 Find Next Closest Hospital	UC-07 Send Request to Hospital	Only happens when the hospital rejects the request (FR-21).

Login is a "Before" condition of the other use cases, so it is not drawn as an include.

Patient

UC-01: Register / Login

Actor	Patient
Other actors	Google Login
Requirements	FR-01, FR-02, FR-03
Before	Patient opens the website.
Steps	1. Patient registers or logs in with email or Google. 2. System opens his home page.
If something goes wrong	Wrong password shows an error. If Google doesn't work, he uses email.
After	Patient is logged in.
Relationships	None.

UC-02: Fill Medical Profile

Actor	Patient
Requirements	FR-05, FR-06, FR-07
Before	Patient is logged in.
Steps	1. Patient writes his allergies, diseases and medicines. 2. He saves it.
If something goes wrong	He can skip it and still send requests.
After	Profile is saved and only staff can see it.
Relationships	None.

UC-03: Add Emergency Contacts

Actor	Patient
Requirements	FR-08, FR-09
Before	Patient is logged in.
Steps	1. Patient adds a name, phone and relation. 2. He saves it.
If something goes wrong	Wrong phone number, the system asks him to fix it.
After	Contact is saved and will get updates.
Relationships	None.

UC-04: Send Emergency Request

Actor	Patient
Other actors	Maps
Requirements	FR-10, FR-11, FR-12, FR-13, FR-14
Before	Patient is logged in.
Steps	1. Patient presses the emergency button. 2. He picks the emergency type. 3. System takes his GPS location and sends the request. 4. System sends a message to his emergency contacts (UC-19).
If something goes wrong	GPS is off, he writes his address himself (UC-20).
After	Request shows on the operator's dashboard as "Submitted".
Relationships	include UC-19 Notify Emergency Contacts. UC-20 Enter Address Manually extends it.

UC-05: Track the Ambulance

Actor	Patient
Other actors	Maps
Requirements	FR-15, FR-16
Before	An ambulance is on the way.
Steps	1. Patient sees the ambulance on the map and the arrival time. 2. The page updates by itself.
If something goes wrong	Live updates stop, the page refreshes every few seconds.
After	Patient knows where the ambulance is.
Relationships	None.

Operator

UC-06: Review Request

Actor	Operator
Requirements	FR-17, FR-18
Before	A new request is waiting.
Steps	1. Operator opens the new request. 2. AI suggests the type and priority. 3. Operator agrees or changes it.
If something goes wrong	AI is off, he picks from a list.
After	Request has a confirmed priority and its status is "Confirmed".
Relationships	None.

UC-07: Send Request to Hospital

Actor	Operator
Other actors	Maps
Requirements	FR-19, FR-20, FR-21
Before	The request was reviewed.
Steps	1. System suggests the closest hospital with the right department and free beds. 2. Operator sends the request to that hospital.
If something goes wrong	The hospital rejects, the system finds the next closest hospital by GPS (UC-21) and the operator sends the request to it.
After	A hospital accepted the request.
Relationships	UC-21 Find Next Closest Hospital extends it.

UC-08: Follow Active Cases

Actor	Operator
Other actors	Emergency Contacts
Requirements	FR-22, FR-23
Before	Operator is logged in.
Steps	1. Operator sees all open cases and their status. 2. He can call the patient or the family if he needs to.
After	He knows the state of every case.
Relationships	None.

Hospital Staff

UC-09: Accept Request and Send Ambulance

Actor	Hospital Staff
Other actors	Maps
Requirements	FR-24, FR-25, FR-26, FR-27
Before	The hospital got a request from the operator.
Steps	1. Hospital staff see the patient's info, the emergency type and the location. 2. They accept the request if they have a free bed and a free paramedic. 3. System suggests the hospital's closest free ambulance. 4. They confirm it and send it.
If something goes wrong	No free beds or no paramedic, so they reject it and the operator sends it to the next closest hospital.
After	Status is "Ambulance Assigned" and the paramedic gets the job.
Relationships	None.

UC-10: Receive Patient and Close Case

Actor	Hospital Staff
Requirements	FR-28, FR-29
Before	The ambulance is going to the hospital.
Steps	1. Staff get ready for the patient. 2. They mark the patient as received when he arrives. 3. When treatment is done, they close the case.
After	The case is closed and its status is "Completed".
Relationships	None.

UC-11: Update Free Beds

Actor	Hospital Staff
Requirements	FR-30
Before	Staff member is logged in.
Steps	1. Staff update the free beds in each department.
After	Hospital suggestions use the new numbers.
Relationships	None.

Paramedic

UC-12: Get the Job

Actor	Paramedic
Other actors	Maps
Requirements	FR-31, FR-32
Before	The hospital sent him to the case.
Steps	1. Paramedic gets the job from the hospital with the location, the emergency type and the patient's allergies and diseases. 2. He goes to the patient.
After	Ambulance is on the way.
Relationships	None.

UC-13: Update Status

Actor	Paramedic
Other actors	Maps
Requirements	FR-33, FR-34, FR-35, FR-36, FR-37
Before	He got the job.
Steps	1. Paramedic updates the status: on the way, arrived, picked up, going to hospital, at hospital. 2. Everyone sees the new status on the map. 3. Emergency contacts get a short update (UC-19).
If something goes wrong	Weak signal, the status is sent when he's back online.
After	Every status is saved in the case history.
Relationships	include UC-19 Notify Emergency Contacts.

Doctor

UC-14: View Case

Actor	Doctor
Requirements	FR-38, FR-39
Before	Patient was received.
Steps	1. Doctor opens the case and the medical profile.
If something goes wrong	No medical profile, he sees the request info only.
After	Doctor has the info he needs.
Relationships	None.

UC-15: Write Notes

Actor	Doctor
Requirements	FR-40
Before	Doctor opened the case.
Steps	1. Doctor writes his first check and treatment notes. 2. He saves them.
After	Notes are saved with the case.
Relationships	None.

Admin

UC-16: Manage Users

Actor	Admin
Requirements	FR-04, FR-41
Before	Admin is logged in.
Steps	1. Admin adds, edits or stops users and gives them roles.
After	Users can log in with the right role.
Relationships	None.

UC-17: Manage Ambulances and Hospitals

Actor	Admin
Requirements	FR-42
Before	Admin is logged in.
Steps	1. Admin adds, edits or removes hospitals and ambulances, and links each ambulance to its hospital.
After	They can be used in suggestions.
Relationships	None.

UC-18: View Reports

Actor	Admin
Requirements	FR-43
Before	Admin is logged in.
Steps	1. Admin picks daily, weekly or monthly. 2. He sees the number of cases and the average response time.
After	Admin sees how the system is doing.
Relationships	None.

Supporting use cases

These three were separated from the steps of other use cases so that include and extend can be shown in the diagram.

UC-19: Notify Emergency Contacts

Actor	Emergency Contacts (they only receive the message)
Requirements	FR-13, FR-35
Before	The patient saved his emergency contacts (UC-03), and a request was sent (UC-04) or the status changed (UC-13).
Steps	1. System sends a message to the patient's emergency contacts when he sends a request. 2. System sends a short update when the case status changes.
After	The emergency contacts know what is happening with the patient.
Relationships	Included by UC-04 Send Emergency Request and UC-13 Update Status.

UC-20: Enter Address Manually

Actor	Patient
Requirements	FR-12
Before	The patient is sending a request (UC-04) and GPS is off.
Steps	1. Patient writes his address himself. 2. System sends the request with this address.
After	The request has a location and shows on the operator's dashboard as "Submitted".
Relationships	Extends UC-04 Send Emergency Request (only when GPS is off).

UC-21: Find Next Closest Hospital

Actor	Operator
Other actors	Maps
Requirements	FR-21
Before	A hospital rejected the request (UC-07, UC-09).
Steps	1. System finds the next closest hospital by GPS. 2. Operator sends the request to it.
After	The request is with a new hospital.
Relationships	Extends UC-07 Send Request to Hospital (only when the hospital rejects).